

Advanced International Macroeconomics B (EI136)

Class 9

Inflation

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April 21, 2026

Outline of today class

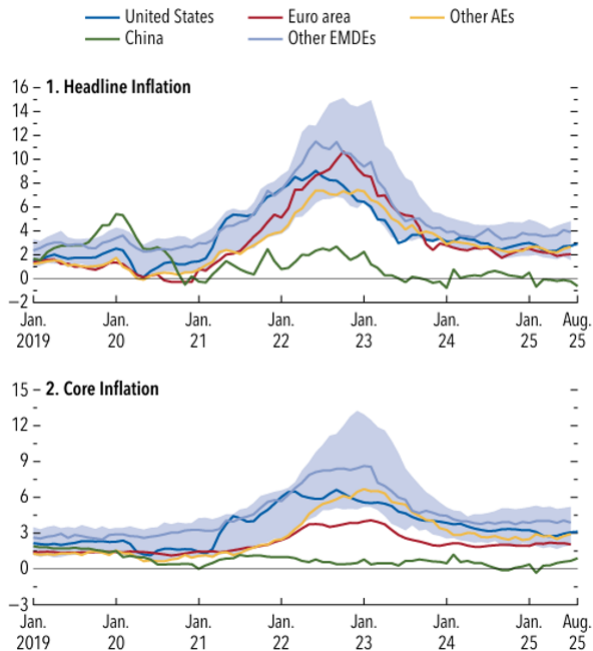
- The recent increase in inflation.
 - A new database with a longer view (Ha, Kose, and Ohnsorge 2023).
- Drivers.
 - Drivers of global inflation (Ha, Kose, Ohnsorge, and Yilmazkuday 2024).
 - Role of supply and demand (Eickmeier and Hofmann 2025, Giannone and Primiceri 2024).
 - Lessons learnt from the recent inflation (IMF 2024).
 - The non-linear Phillips curve (Benigno and Eggertsson 2024, Beaudry, Hou and Portier 2025).
- Understanding expectations (IMF 2023, Coibion and Gorodnichenko 2025).

PATTERN OF INFLATION

- Inflation surged, came down rapidly.

Figure 1.6. Global Inflation Trends

(Percent, year over year)



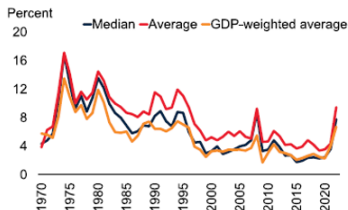
IMF (2025), World Economic Outlook, October.

- Database of several inflation indicators for a broad range of countries (Ha, Kose, and Ohnsorge 2023).
 - <https://www.worldbank.org/en/research/brief/inflation-database>
 - 38 advanced countries and 171 emerging ones, since 1970, also advanced and emerging median.
 - Several indicators (not for all countries): headline inflation, core, energy and food, PPI, GDP deflator, global commodity prices.
- Broad inflation movements seen across the world.
 - Increase with the 1970's oil shocks.
 - Policy tightening brought inflation down in advanced economies in the 1980's, decrease also in emerging economies in 1990's.
 - Low inflation since 2000, until recent increase.

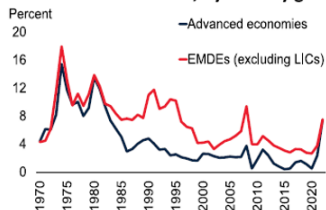
Inflation since 1970

- Surge in the 1970's, decrease until 2000, recent increase.

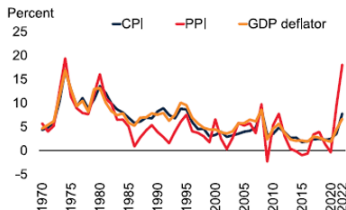
A. Global CPI inflation



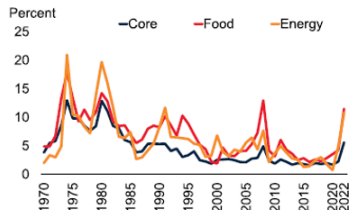
B. Headline CPI inflation, by country group



C. Global headline CPI, PPI inflation, and the GDP deflator



D. Global core, food, and energy inflation



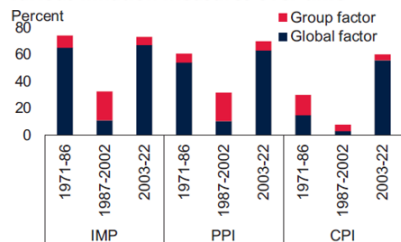
Ha, Jongrim, Ayhan Kose, and Franziska Ohnsorge (2023). "One-Stop Source: A Global Database of Inflation", *Journal of International Money and Finance* 137.

- Split inflation into a global factor, a group factor (Adv and Eme), and the residual.
 - Contrast across inflation measures with different tradable components.
- Global component more important for import prices and PPI than for CPI.
- Global component decreased from mid 1980's to early 2000's, but has since increased.
 - High recent value also for CPI.

Factor decomposition

- Strong global component in the recent episode.

A. Contribution of inflation factors from various inflation measures over time



B. Contribution of inflation factors from various inflation measures over time

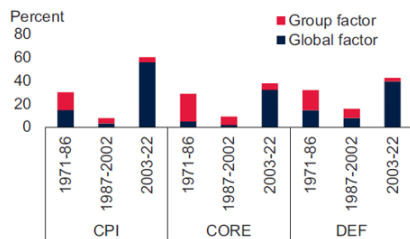


Fig. 3. Inflation synchronization: different measures of inflation Sources: World Bank. Notes: The global inflation factors are estimated with a baseline dynamic factor model (two-factor model) for annual inflation in 38 countries (25 advanced economies and 13 EMEs) for the period of 1971–2022. “IMP”, “PPI”, “CPI”, “CORE”, “DEF” indicate inflation in import prices, producer price index, headline CPI, core CPI, and GDP deflator, respectively. Long-term trend is eliminated using 15-year moving average. All numbers refer to sample medians.

Ha, Jongrim, Ayhan Kose, and Franziska Ohnsorge (2023). “One-Stop Source: A Global Database of Inflation”, *Journal of International Money and Finance* 137.

DRIVERS OF GLOBAL INFLATION

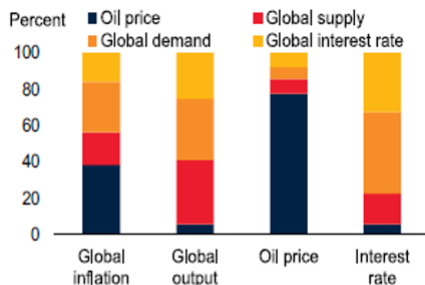
Principal component of inflation

- Assess the drivers of the principal component across G7 countries since 1970 (Ha, Kose, Ohnsorge, and Yilmazkuday 2024).
- Extract principal components of inflation, output and interest rates, and run a VAR with sign restrictions to extract shocks.
 - Supply, demand, interest rate, and oil price shocks.
- Since 1970, oil price and demand shocks drive inflation (38% and 28% of variance at 2 years horizon).
 - Rising role of oil price in last 20 years.
- Some sensitivity to inflation measure: oil prices play little role for core CPI (as expected), which is more sensitive to supply (41%) and demand and interest (26% each).
- Evidence for better grounded policy, as inflation has become less sensitive to shocks.
- In the 2020 episode, demand (fall in Covid, and subsequent recovery) was the driver of inflation., with oil price also playing a role after the recession.

Variance decomposition

- Prominent role of demand and oil price shocks.

A. Variance decompositions of global variables: 1970-2022



B. Variance decompositions of global inflation over time

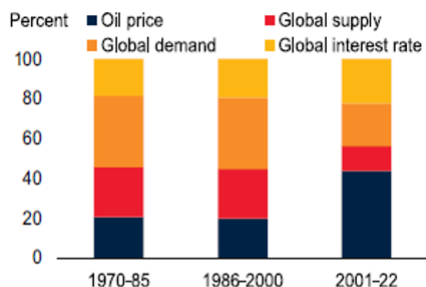


Fig. 3 Contributions of global shocks to global inflation variation. *Notes:* The figures present variance decompositions of global variables (A) and global inflation (B) at the 2-year forecasting horizons. The results are medians of 1000 successful Bayesian draws based on the global FAVAR model.

Sensitivity of inflation

- More moderate impulse response to demand, supply and interest rate shock in recent years. Little change to oil shocks.

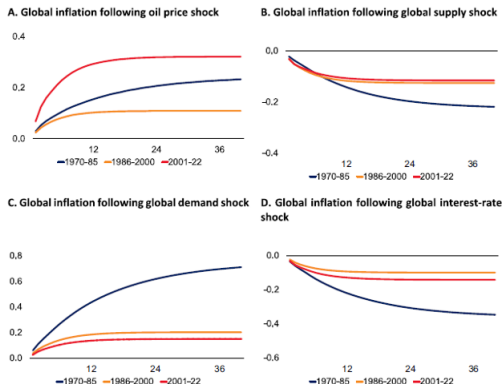


Fig.5 Cumulative impulse responses of global inflation over time. *Notes:* The impulse responses (in percentage point) are based on three sub-sample global FAVAR estimations (1970–1985, 1986–2000, 2001–2022). Three solid lines with different colors indicate median responses to a one-standard-deviation global shock for the corresponding three sub-sample periods. Horizontal axis indicates month(s).

Ha, Jongrim, Avhan Kose, Franziska Ohnsorge, and Hakan Yilmazkuday (2024). "What Explains Global Inflation" *IMF Economic Review*

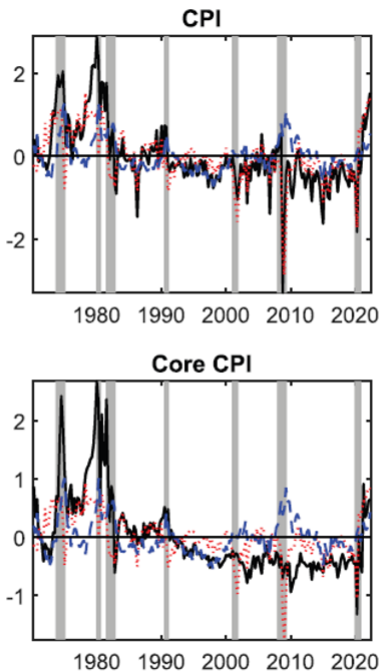
DEMAND AND SUPPLY

Demand or supply?

- Identify broad driving sources of inflation in US and Euro area (Eickmeier and Hofmann, 2025).
- Broad range of inflation and real activity indicators (140 quarterly since 1970).
- Factor model, 3 factors explain 52% of variance.
- Putting economic interpretation on factors.
 - Factor analysis only gives combination of factors and individual loadings on them (not factor and loadings separately).
 - Sign restrictions: the demand factor raises both inflation and real activity, the supply factor raises activity and reduces inflation.
 - Random draws of combination (“rotation”), keep the ones satisfying the restrictions (200, and median factor).
- During 2008 crisis, both demand and supply down with offsetting effects on inflation.
- Since Covid, demand very positive and supply very negative, both driving inflation.

- Recent inflation in the US primarily demand driven (but not only).

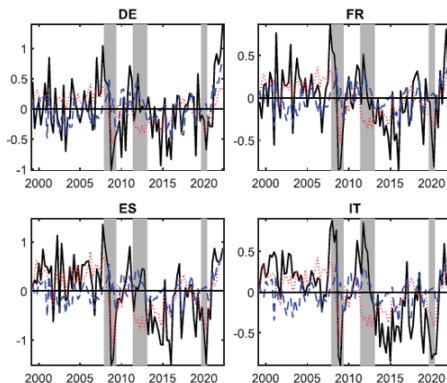
Figure 3. Historical Decompositions



Eickmeier, Sandra, and Boris Hofmann (2025) "What Drives Inflation? Disentangling Demand and Supply Factors", *International Journal of Central Banking* 21(3), pp. 1-44

- 4 countries, larger role for supply since 2021 than in the US.

Figure 10. Historical Decompositions of Euro-Area CPI Inflation



Note: Quarter-on-quarter, in percent. Solid black: demeaned time series. Dotted red: contributions of the Median Target demand factor. Dashed blue: contributions of the Median Target supply factor. Gray bars: CEPR recessions.

Eickmeier, Sandra, and Boris Hofmann (2025) "What Drives Inflation? Disentangling Demand and Supply Factors", *International Journal of Central Banking* 21(3), pp. 1-44

- Re-assessment of supply and demand factors (Giannone and Primiceri 2024).
 - Sign restrictions in VAR.
- If aggregate demand is flat, supply shocks affect output but only moderately inflation. Demand shifts affect inflation.
- Contrast US and Europe: similar inflation, more persistent lower GDP in Europe.
 - Supply reduced economic activity, but demand boost raised it and was inflationary.
 - Robust to extending to oil prices.

Decomposition of drivers

- Growth reflects demand and supply, but inflation more from the demand side.

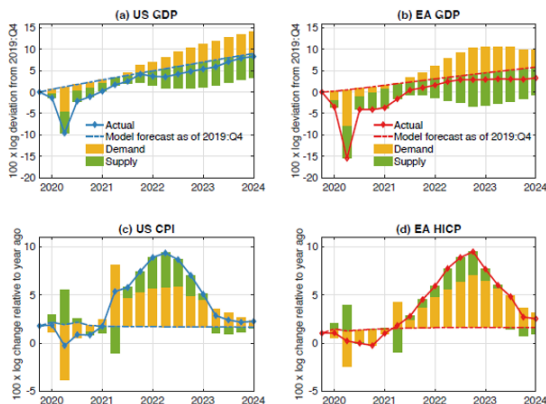


Figure 5: Historical decomposition of GDP and inflation dynamics. Sources: Data from Eurostat, the European Central Bank, the Bureau of Labor Statistics, and the Bureau of Economic Analysis; accessed via Haver Analytics; computations by authors.

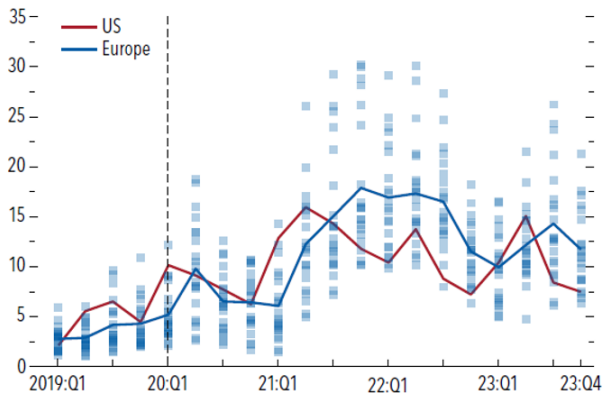
Tacking stock of the recent experience

- Assess drivers of inflation and new elements highlighted in recent years (IMF 2024).
- Supply shocks, and bottlenecks from changing demand (shift from services to goods during the pandemic).
 - High dispersion of inflation across sectors.
- Non-linear Phillips curve, steeper when bottlenecks arise.
- Granual sectoral model with input-ouput relation (bottleneck is a supply shock for sectors down the chain).
- Rising demand increases the share of sectors with bottlenecks, leading to steeper Phillips curve.
- Bottlenecks are temporary, so policy can “look through” to some extent.

Rising sectoral dispersion

- Dispersion of inflation across sectors points to bottlenecks.

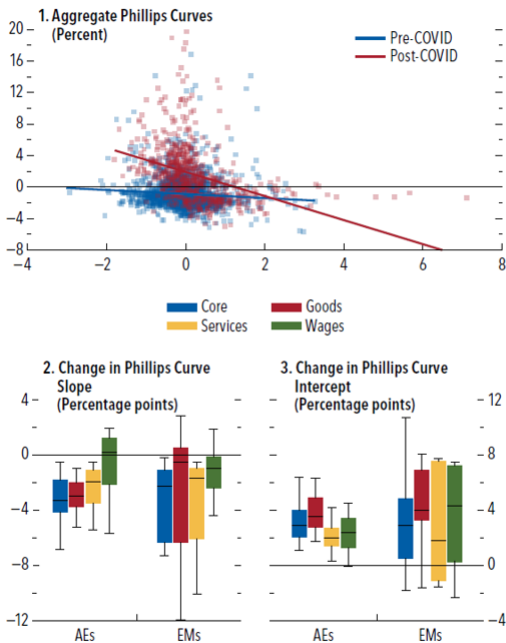
Figure 2.2. Movements in Sectoral Price Dispersion
(Percent)



IMF (2024). "The Great Tightening: Insights from the Recent Inflation Episode", World Economic Outlook, October, chapter 2

- Steepening of the Phillips curve (more negative slope).

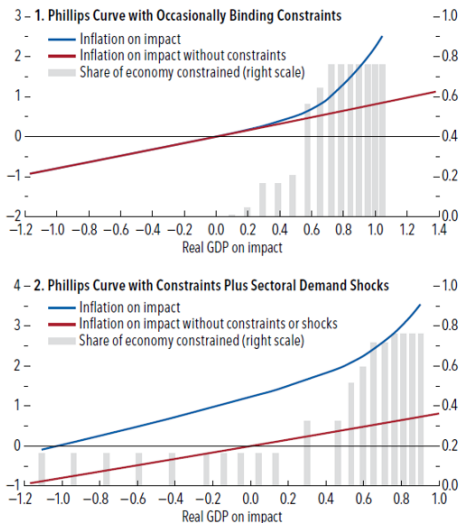
Figure 2.6. Evolution of Phillips Curves



IMF (2024). "The Great Tightening: Insights from the Recent Inflation Episode", World Economic Outlook, October, chapter 2

- Higher growth (towards the right) increase share of sectors with bottlenecks, and make the Phillips curve steeper.

Figure 2.11. Phillips Curve under Different Constraints
(Percent)



Source: IMF staff calculations.

Note: The blue line shows combination of the impact effect of real GDP (x-axis) and inflation (left scale) on monetary policy shocks of various sizes, with panel 2 also including a relative demand shock. The gray bars (right scale) show the share of the economy constrained. The red line shows the same combination without any supply bottlenecks imposed. The Phillips curve shape will depend on choice of constraints.

IMF (2024). "The Great Tightening: Insights from the Recent Inflation Episode", World Economic Outlook, October, chapter 2

NON – LINEAR PHILLIPS CURVE

Usual curve, and recent puzzles

- New Keynesian Phillips Curve has inflation π_t driven by slack (x_t), expected inflation ($E_t\pi_{t+1}$), and cost-push shocks (μ_t).

$$\pi_t = \kappa x_t + \beta E_t \pi_{t+1} + \mu_t$$

- Usual estimates point to a limited slope κ (flat Phillips curve).
- Recent inflation increase took central banks (and economists) by surprise.
 - Inflation expectations remained steady, so it was expected that inflation would be temporary.
- NKPC ignores non-linearities, which may be very relevant (Benigno and Eggertsson 2024).
 - Labor market tightness, measures by ratio of vacancies to unemployment, θ , was unusually high.
 - Evidence that curve is steeper when $\theta > 1$.

Non-linear curve

- Curve gets steeper when tightness θ is higher.

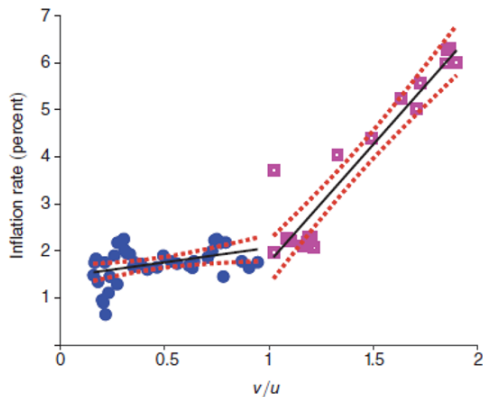


FIGURE 1. UNITED STATES 2009–2023: CPI INFLATION RATE AND VACANCY-TO-UNEMPLOYED RATIO (v/u)

Benigno, Pierpaolo, and Gauti Eggertsson (2024). "The Slanted-L Phillips Curve", *AEA Papers and Proceedings* 114, pp. 84-89.

Multi country evidence

- Slanted L Phillips curve: steep when unemployment below a threshold.

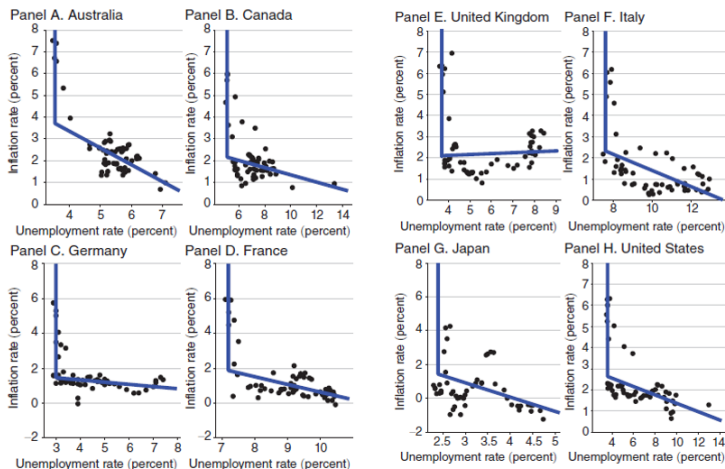


FIGURE 2. INTERNATIONAL EVIDENCE ON INFLATION
AND UNEMPLOYMENT TRADE-OFF, 2009–2023

Benigno, Pierpaolo, and Gauti Eggertsson (2024). "The Slanted-L Phillips Curve", *AEA Papers and Proceedings* 114, pp. 84-89.

Controlling for expectations

- New Keynesian Phillips curve shows that inflation reacts to a) output gap (possibly linearly) and b) inflation expectations.
- Problem: inflation expectations are high with the labor market is tight. Beware of omitted variable bias (Beaudry, Hou and Portier 2025).
- Use cross sectional data of US metropolitan areas.
 - Time fixed-effect to capture inflation expectations.
 - Can refine with area-level expectation measures.
- The Phillips curve is linear once we control for expectations.
 - Sensitivity of measure: non-linearity disappears using households' expectations, but not so much using forecasters' expectations.

Fixed effect

- Non linearity at metropolitan level disappears with time fixed effects.

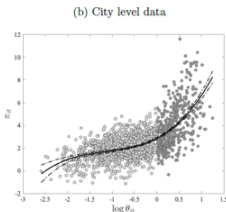
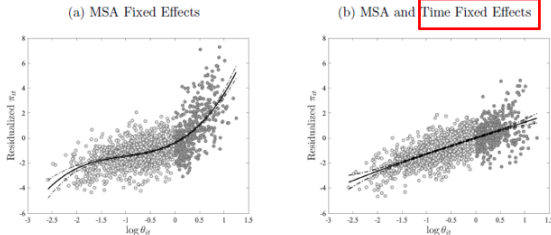


Figure 3: Inflation and Labour Market Tightness, City Level Data



Beaudry, Paul, Chenyu Hou, and Franck Portier (2025). "On the Fragility of the Nonlinear Phillips Curve View of Recent Inflation", CEPR discussion paper 19950.

Regression analysis

- Time fixed effects remove the non-linearity

Table 2: Estimation of Iterated Forward (2) with OLS

	Quarter-to-quarter				Year-to-year			
	Core CPI	HL CPI	Core CPI	HL CPI	Core CPI	HL CPI	Core CPI	HL CPI
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
β_θ	1.19*** (0.145)	1.61*** (0.328)	1.13*** (0.209)	1.03*** (0.328)	0.89*** (0.071)	1.24*** (0.154)	1.08*** (0.092)	0.86*** (0.137)
$\beta_{D \times \theta}$	2.86*** (0.457)	-0.43 (0.586)	3.71*** (0.665)	-0.15 (0.586)	3.81*** (0.220)	0.03 (0.270)	5.11*** (0.288)	0.08 (0.241)
β_D	-0.18 (0.241)	-0.21 (0.252)	0.06 (0.349)	-0.16 (0.253)	-0.27** (0.118)	-0.23** (0.118)	-0.24 (0.154)	-0.23** (0.106)
$p_{i,t-1}$	-0.00 (0.001)	-0.01*** (0.005)	-0.01*** (0.002)	-0.02*** (0.006)				
$p_{i,t-4}$					-0.00* (0.003)	-0.04*** (0.010)	-0.03*** (0.003)	-0.05*** (0.010)
Observations	1555	1555	1555	1555	1492	1492	1492	1492
MSA F.E.	Y	Y	Y	Y	Y	Y	Y	Y
Time F.E.	N	Y	N	Y	N	Y	N	Y

Notes: We estimate $\pi_{i,t} = \beta_\pi \pi_{i,t-1}^e + \beta_\theta \log \theta_{i,t} + \beta_{D \times \theta} D_{i,t} \times \log \theta_{i,t} + \beta_D D_{i,t} + \beta_p p_{i,t-1} + \alpha_i + X_t + \varepsilon_{i,t}$

Beaudry, Paul, Chenyu Hou, and Franck Portier (2025). "On the Fragility of the Nonlinear Phillips Curve View of Recent Inflation", CEPR discussion paper 19950.

Controlling for expectations

- Linear Phillips curve when controlling for one-quarter expectations

Table 4: Estimation of Equation (2) with Local Expectations

	Quarter-to-quarter Core CPI					
	(1)	(2)	(3)	(4)	(5)	(6)
β_{π^e}			0.76*** (0.072)			
β_θ	1.86*** (0.336)	1.85*** (0.640)	1.85*** (0.313)	1.85*** (0.315)	2.04*** (0.667)	1.34 (1.067)
$\beta_{D \times \theta}$	2.09*** (0.628)	-0.68 (0.758)	-0.30 (0.627)	-1.02* (0.588)	-1.23 (0.790)	3.52 (4.177)
β_D	-0.44 (0.310)	-0.09 (0.297)	-0.19 (0.289)	-0.11 (0.290)	0.17 (0.310)	-0.70* (0.420)
$p_{i,t-1}$	0.00 (0.004)	-0.02 (0.010)	-0.00 (0.004)	-0.00 (0.004)	-0.02* (0.010)	-0.02 (0.011)
Observations	741	741	741	741	741	741
MSA F.E.	Y	Y	Y	Y	Y	Y
Time F.E.	N	Y	N	N	Y	N

Notes: We estimate $\pi_{i,t} = \beta_{\pi^e} \pi_{i,t+1}^e + \beta_\theta \log \theta_{i,t} + \beta_{D \times \theta} D_{i,t} \times \log \theta_{i,t} + \beta_D D_{i,t} + \beta_p p_{i,t-1} + \alpha_i + \Theta_t + \varepsilon_{i,t}$. Column (1) and (2) are estimations without $\pi_{i,t+1}^e$. Column (3) reports results with MSA level $\pi_{i,t+1}^e$ but without time fixed effects. Columns (4)-(6) use $\pi_t - 0.99 \times \pi_{i,t+1}^e$ as dependent variable. Column (6) is instrumenting for $\log \theta$ and the interaction term with shift-share IV and its product with $D_{i,t}$. Sample is 2013Q3-2024Q3.

Beaudry, Paul, Chanyu Hou, and Franck Portier (2025). "On the Fragility of the Nonlinear Phillips Curve View of Recent Inflation", CEPR discussion paper 19950.

- Expectations remove the non-linearity, especially from household survey.

Table 5: Estimation of Equation (3), OLS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
β_{π^e}	0.86*** (0.08)	0.84*** (0.08)	0.91*** (0.08)	0.71*** (0.10)	0.98*** (0.13)	1.08*** (0.14)	1.09*** (0.14)	0.71*** (0.14)
β_θ	0.08 (0.17)	0.12 (0.17)	-0.07 (0.18)	-0.03 (0.17)	0.74*** (0.19)	0.77*** (0.18)	0.74*** (0.19)	0.47*** (0.17)
$\beta_{\theta \times D}$	4.05*** (0.65)	4.00*** (0.64)	2.38*** (0.87)	1.26 (0.94)	0.01 (0.84)	-0.25 (0.83)	-0.56 (1.15)	-1.84* (1.04)
β_v		-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)		-0.05*** (0.02)	-0.05*** (0.02)	-0.04** (0.02)
$\beta_{v \times D}$		0.14*** (0.05)	0.17*** (0.05)	0.14*** (0.05)		0.10* (0.06)	0.10* (0.06)	0.08 (0.05)
β_D			0.99*** (0.37)	0.96*** (0.36)			0.17 (0.42)	0.50 (0.37)
$\beta_{\pi_{-1}}$				0.23*** (0.08)				0.45*** (0.07)
N	144	144	144	144	144	144	144	144
adj. R^2	0.60	0.62	0.64	0.65	0.48	0.51	0.50	0.61

Notes: The generic Phillips curve we estimate is $\pi_t = \beta_{\pi^e} \pi_{t+1}^e + \beta_\theta \log \theta_t + \beta_{D \times \theta} D_t \times \log \theta_t + \beta_v v_t + \beta_{D \times v} D_t \times v_t + \beta_{\pi_{-1}} \pi_{t-1} + \beta_D D_t + \varepsilon_t$. Inflation is quarter-to-quarter CPI core inflation (annualized). D is a dummy for $\log \theta_t > 0$ and v is a measure of supply shocks. Columns (1) to (4) use the Federal Reserve Bank of Cleveland measure of inflation expectations. Columns (5) to (8) use the Michigan Survey of Consumers measure of inflation expectations. Both measures are one year-ahead and adjusted to obtain one quarter-ahead expectation (see Appendix D). Sample is 1988Q1-2023Q4.

Beaudry, Paul, Chenuv Hou, and Franck Portier (2025). "On the Fragility of the Nonlinear Phillips Curve View of Recent Inflation", CEPR discussion paper 19950.

INFLATION EXPECTATIONS

Recent behavior of expectations

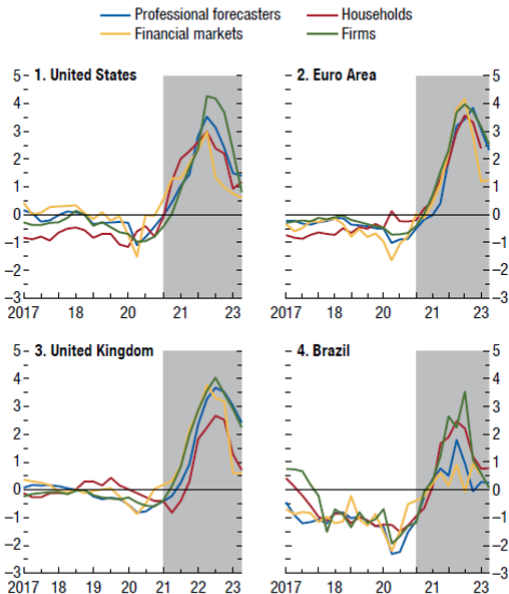
- Measures of expectations for households, from financial markets (indexed bonds), professional forecasters, and firms (using earning calls). IMF 2023.
- Increase in all measures of expectations in the recent episode.
- Increase concentrated in expectation over next 12 months. Expectations are better anchored at longer horizons.
- Historical evidence shows that when expectations increase, they take time to come down.

- Broad increase in expectations for the next 12 months.

Figure 2.2. Next-12-Months Mean Inflation Expectations by Economic Agent

(z-score, standard deviations from the mean)

Economic agents agree on the broad dynamics of near-term inflation expectations. The sharp increases in 2022 were unusual compared to the experience of the last 20 years.



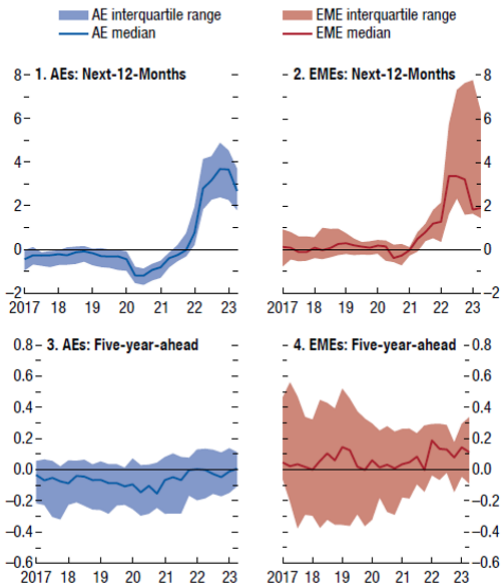
International Monetary Fund (2023) "Managing Expectations: Inflation and Monetary Policy", World Economic Outlook chapter 2, October.

- Longer term expectations are more strongly anchored.

Figure 2.3. Cross-Economy Distribution of Mean Inflation Expectations over Time

(Percentage point deviation from target)

Near-term inflation expectations shot up rapidly from 2022 but are now reverting, while long-term inflation expectations have moved only marginally, but in a narrowing range.



International Monetary Fund (2023) "Managing Expectations: Inflation and Monetary Policy", World Economic Outlook chapter 2, October.

- Hybrid Phillips curve of inflation on expectations and lagged inflation.
- Backward looking component larger for emerging economies.
 - IV approach reduces the impact of expectations (instrument: lagged expectations and output gap).
- Expectations initially played a secondary role in the current increase, but now have a larger impact.
 - Expectations have a larger impact on actual inflation when inflation is high to start with.
- Disinflation policy is easier when expectations are forward looking rather than backward looking.
 - Less persistence of inflation, smaller sacrifice ratio.

- Expectations affect actual inflation.

Figure 2.6. Key Coefficients of the Hybrid Phillips Curve
(Regression coefficients)

Near-term inflation expectations play a larger role in explaining current inflation in advanced economies than in emerging market economies.

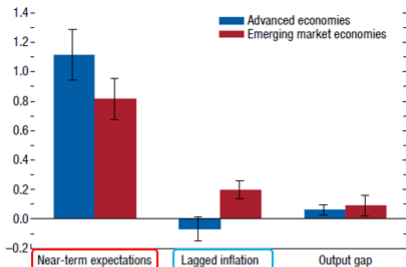
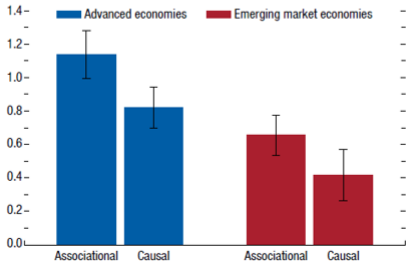


Figure 2.7. Associational versus Causal Estimated Effects of Inflation Expectations on Current Inflation
(Regression coefficients)

Accounting for the influence of current inflation on expectations of future inflation in the Phillips curve reduces the estimated effects of inflation expectations on current inflation by about 30 percent across economy groups.



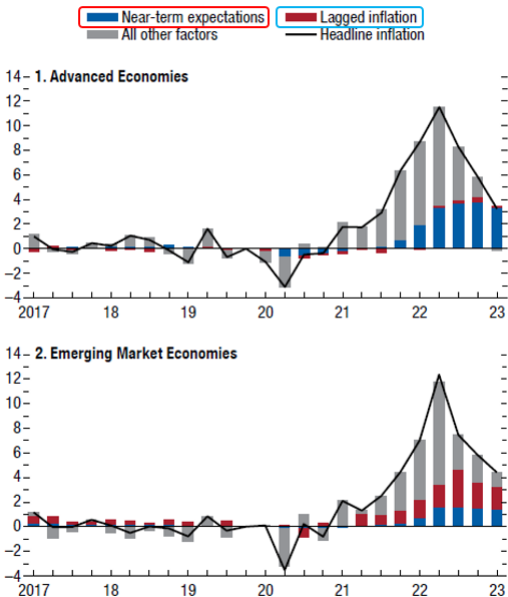
International Monetary Fund (2023) "Managing Expectations: Inflation and Monetary Policy", World Economic Outlook chapter 2, October.

- Growing role of expectations in driving inflation.

Figure 2.8. Contributors to Recent Inflation Dynamics

(Percentage point deviation from 2019:Q4)

A decomposition of the recent dynamics of headline inflation reveals the growing importance of near-term inflation expectations.



International Monetary Fund (2023) "Managing Expectations: Inflation and Monetary Policy", World Economic Outlook chapter 2, October.

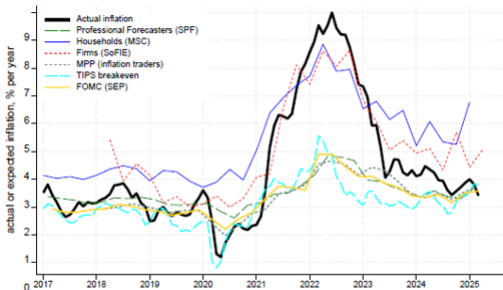
A more skeptical view

- Short term inflation are not anchored, and they are the ones that matter for the Phillips curve (Coibion and Gorodnichenko 2025).
 - They have come down, but not all the way (especially for households).
 - Large role of salient prices, such as gasoline.
- Evidence on sensitivity to additional information, i.e. whether agents are initially well informed.
 - Information reduces mis-perceptions, but not at high inflation as agents are then well informed.
- Households expectations have the strongest impact in the Phillips curve. No evidence of non-linearity
- If expectations remain high, another shock can fuel inflation.

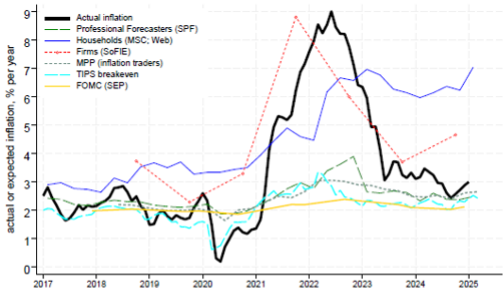
- Volatile expectations, with heterogenous return to normal.

Figure 1: Inflation Expectations Before, During and Since the Surge

Panel A: Mean Year-Ahead Inflation Expectations



Panel B: Mean 5-Year Ahead Inflation Expectations



Notes: inflation expectations are from the Michigan Survey of Consumers (MSC), Survey of Firms' Inflation Expectations (SoFIE), MacroPolicy Perspectives (MPP), the spread between nominal and inflation indexed bonds (TIPS) from Gurkaynak et al. (2010), and Summary of Economic Projections (SEP) from the Federal Open Market Committee (FOMC). To ensure consistency over time, 5-year ahead inflation expectations for MSC are from the subset of response that were collected using an online survey.

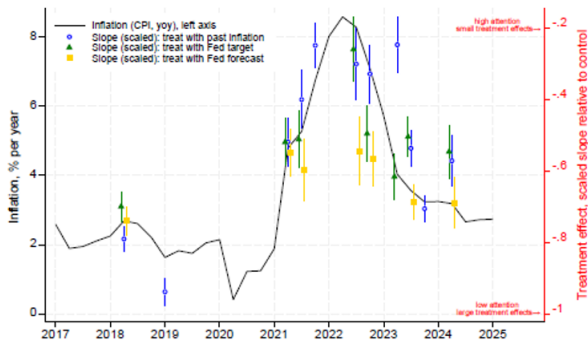
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Attention to information

- Providing information has less effect (dots / lines closer to top) when inflation is high.

Figure 5: How Inattention Varies with the Inflation Environment

Panel A: Information Treatments and the Level of Inflation in the U.S.



Coibion, Olivier, and Yuriy Gorodnichenko (2025). "Inflation, Expectations and Monetary Policy: What Have We Learned and to What End?", CEPR discussion paper 20274

Phillips curve

- Stronger role (coefficient and R2) for **households** expectations than for professional forecasters.

Table 2: Whose Expectations Matter for the Phillips Curve?

Dep. Var.: π_t	(1)	(2)	(3)	(4)	(5)	(6)	(7)
MSC: $E^{1yr} \pi$	1.504*** (0.088)				1.268*** (0.236)	1.574*** (0.584)	1.318*** (0.300)
SPF: $E^{1yr} \pi$		0.957*** (0.133)			0.277 (0.169)		0.299* (0.177)
SoFIE: $E^{1yr} \pi$			0.901*** (0.268)			-0.286 (0.522)	
π_{t-1}				0.687*** (0.102)			-0.045 (0.102)
$U - U^*$	-0.194** (0.079)	-0.232** (0.105)	-0.180 (0.274)	-0.142* (0.073)	-0.208** (0.081)	-0.191 (0.231)	-0.217** (0.094)
Sample	1978-2025	1981-2025	2018-2025	1978-2025	1981-2025	2018-2025	1981-2025
Observations	189	175	28	189	175	28	175
R-squared	0.717	0.267	0.414	0.497	0.487	0.563	0.488

Coibion, Olivier, and Yuriy Gorodnichenko (2025). "Inflation, Expectations and Monetary Policy: What Have We Learned and to What End?", CEPR discussion paper 20274